

INTELLIGENT PATIENT FLOW AND APPOINTMENT SCHEDULING SYSTEM FOR HEALTHCARE FACILITIES

1. Appointment Scheduling Module

- Online appointment booking
- Doctor selection
- Specialty selection
- Appointment rescheduling
- Appointment cancellation

2. Queue Management Module

- Automatic queue number generation
- Queue prioritization
- Real-time queue tracking
- Service point allocation

3. Doctor Schedule Management Module

- Doctor availability management
- Shift scheduling
- Leave management
- Consultation room allocation

4. Mobile Application and Patient Check-In Module

- Book appointments.
- Monitor queue status.
- Receive notifications.
- View appointment history.
- Self-service check-in
- QR code check-in
- Reception-based check-in

5. Reporting and Analytics Module

- Patient waiting time reports
- Appointment utilization reports
- Doctor workload analysis
- Daily attendance reports

6. AI-Based Queue Prediction

- Predict waiting times.
- Forecast patient arrivals.
- Suggest optimal appointment slots.